



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 06/06/2024*

EXERCISE

We observe N samples of the following w.s.s. discrete-time real random process:

$$X[n] = S[n] + W[n] = A \sin(2\pi f_0 n + \theta) + W[n], \quad n = 0, 1, \dots, N-1,$$

where A is the amplitude of the Signal of Interest (SOI) and it is modelled as a Rayleigh distributed random parameter with power $E\{A^2\} = \sigma_A^2$, the initial phase θ is modelled as a random parameter uniformly distributed in $[-\pi, \pi)$, $W[n]$ is additive white Gaussian noise (AWGN) with Power Spectral Density (PSD) $S_w(e^{j2\pi f}) = \sigma_w^2$. The amplitude A , the phase θ , and the noise samples are mutually independent. The digital frequency f_0 is a-priori known deterministic.

(1) Derive the Autocorrelation Function (ACF) and the Power Spectral Density (PSD) of the w.s.s. process $X[n]$ and plot them.

(2) Derive the Linear Minimum Mean Square Error (LMMSE) estimator of the signal of interest (SOI) at time $n+1$, $S[n+1]$, given the observation of the two samples $X[n]$ and $X[n-1]$. Express

$$\hat{S}[n+1] \text{ as a function of the signal to noise ratio (SNR), } \gamma = \frac{E\{S^2[n]\}}{E\{W^2[n]\}} = \frac{\sigma_A^2}{2\sigma_w^2}.$$

(3) Derive the bias and the Mean Square Error (MSE) σ_ϵ^2 of the LMMSE estimator of $S[n+1]$ and express it as a function of γ .

(4) Derive $\hat{S}[n+1]$ and σ_ϵ^2 when $\gamma \rightarrow 0$ and $\gamma \rightarrow \infty$, and comment on the results. Calculate the estimator and its MSE numerically when $\sigma_A^2 = 1$, $\gamma = 2$, and $f_0 = 1/6$.

Hint: $\sin(\alpha)\sin(\beta) = \frac{\cos(\alpha - \beta) - \cos(\alpha + \beta)}{2}$, $\cos^2(\alpha) = \frac{1 + \cos(2\alpha)}{2}$, $\sin^2(\alpha) = \frac{1 - \cos(2\alpha)}{2}$



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Solution

(1) ACF:

$$X[n] = S[n] + W[n] = A \sin(2\pi f_0 n + \theta) + W[n], \quad n = 0, 1, \dots, N-1$$

$$R_x[m] = R_s[m] + R_w[m] = R_s[m] + \sigma_w^2 \delta[m]$$

$$\begin{aligned} R_s[m] &= E\{A \sin(2\pi f_0 n + \theta) \cdot A \sin(2\pi f_0 (n+m) + \theta)\} \\ &= E\{A^2\} E\{\sin(2\pi f_0 n + \theta) \sin(2\pi f_0 (n+m) + \theta)\} \\ &= \sigma_A^2 E\left\{\frac{\cos(2\pi f_0 m) - \cos(2\pi f_0 (2n+m) + 2\theta)}{2}\right\} \\ &= \frac{\sigma_A^2}{2} \cos(2\pi f_0 m) + \frac{\sigma_A^2}{2} E\{\cos(2\pi f_0 (2n+m) + 2\theta)\} \\ &= \frac{\sigma_A^2}{2} \cos(2\pi f_0 m) \\ R_x[m] &= \frac{\sigma_A^2}{2} \cos(2\pi f_0 m) + \sigma_w^2 \delta[m] \end{aligned}$$

PSD:

$$S_x(e^{j2\pi f}) = FT\{R_x[m]\} = 2\pi \frac{\sigma_A^2}{4} \delta(e^{j2\pi f} - e^{j2\pi f_0}) + 2\pi \frac{\sigma_A^2}{4} \delta(e^{j2\pi f} - e^{-j2\pi f_0}) + \sigma_w^2$$

(2) LMMSE estimator of A .

$$\hat{S}[n+1] = a_1 X[n] + a_2 X[n-1] + b, \quad E\{S[n]\} = 0, \quad E\{X[n]\} = 0 \rightarrow b = 0 \rightarrow \hat{S}[n+1] = \mathbf{a}^T \mathbf{X}$$

By applying the Orthogonality principle, we find that the coefficients of the LMMSE estimator are given by $\mathbf{a} = \mathbf{C}_x^{-1} \mathbf{c}_{xs}$, where $\mathbf{C}_x = \mathbf{C}_s + \mathbf{C}_w = \mathbf{C}_s + \sigma_w^2 \mathbf{I}$. In our case, the elements of the signal

covariance matrix are: $[\mathbf{C}_s]_{n,n+m} = R_s[m] = \frac{\sigma_A^2}{2} \cos(2\pi f_0 m)$.

The 2D vector of the cross-covariances between the signal we want to estimate and the observed data is given by:

$$\mathbf{c}_{xs} = E\{\mathbf{X}S[n+1]\} = E\left\{\begin{bmatrix} X[n]S[n+1] \\ X[n-1]S[n+1] \end{bmatrix}\right\} = \begin{bmatrix} R_{xs}[1] \\ R_{xs}[2] \end{bmatrix} = \begin{bmatrix} R_s[1] \\ R_s[2] \end{bmatrix} = \frac{\sigma_A^2}{2} \begin{bmatrix} \cos(2\pi f_0) \\ \cos(4\pi f_0) \end{bmatrix}$$



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As a consequence, the LMMSE estimator of $S[n+1]$ is given by:

$$\begin{aligned}
 \hat{S}[n+1] &= a_1 X[n] + a_2 X[n-1] \\
 &= \frac{\sigma_A^2}{2} \begin{bmatrix} \cos(2\pi f_0) \\ \cos(4\pi f_0) \end{bmatrix}^T \begin{bmatrix} \frac{\sigma_A^2}{2} + \sigma_w^2 & \frac{\sigma_A^2}{2} \cos(2\pi f_0) \\ \frac{\sigma_A^2}{2} \cos(2\pi f_0) & \frac{\sigma_A^2}{2} + \sigma_w^2 \end{bmatrix}^{-1} \begin{bmatrix} X[n] \\ X[n-1] \end{bmatrix} \\
 &= \gamma \begin{bmatrix} \cos(2\pi f_0) \\ \cos(4\pi f_0) \end{bmatrix}^T \begin{bmatrix} 1+\gamma & \gamma \cos(2\pi f_0) \\ \gamma \cos(2\pi f_0) & 1+\gamma \end{bmatrix}^{-1} \begin{bmatrix} X[n] \\ X[n-1] \end{bmatrix} \\
 &= \frac{\gamma}{(1+\gamma)^2 - \gamma^2 \cos^2(2\pi f_0)} \begin{bmatrix} \cos(2\pi f_0) & \cos(4\pi f_0) \end{bmatrix} \begin{bmatrix} 1+\gamma & -\gamma \cos(2\pi f_0) \\ -\gamma \cos(2\pi f_0) & 1+\gamma \end{bmatrix} \begin{bmatrix} X[n] \\ X[n-1] \end{bmatrix} \\
 &= \frac{\gamma}{(1+\gamma)^2 - \gamma^2 \cos^2(2\pi f_0)} \times \\
 &\quad \times \begin{bmatrix} (1+\gamma) \cos(2\pi f_0) - \gamma \cos(4\pi f_0) \cos(2\pi f_0) & -\gamma \cos^2(2\pi f_0) + (1+\gamma) \cos(4\pi f_0) \end{bmatrix} \begin{bmatrix} X[n] \\ X[n-1] \end{bmatrix} \\
 &= \gamma \cdot \frac{(1+\gamma) \cos(2\pi f_0) - \gamma \cos(4\pi f_0) \cos(2\pi f_0)}{(1+\gamma)^2 - \gamma^2 \cos^2(2\pi f_0)} X[n] - \gamma \cdot \frac{\gamma \cos^2(2\pi f_0) - (1+\gamma) \cos(4\pi f_0)}{(1+\gamma)^2 - \gamma^2 \cos^2(2\pi f_0)} X[n-1]
 \end{aligned}$$

where: $\gamma \triangleq \frac{\sigma_A^2}{2\sigma_w^2}$

(3) Bias and MSE of the LMMSE estimator of A .

$$E\{\varepsilon\} = E\{S[n+1] - \hat{S}[n+1]\} = E\{S[n+1]\} - E\{\hat{S}[n+1]\} = 0 \rightarrow \hat{S}[n+1] \text{ is unbiased}$$



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$$\begin{aligned}
 MSE\{\hat{S}[n+1]\} &= E\left\{\left(S[n+1] - \hat{S}[n+1]\right)^2\right\} = E\left\{S[n+1]\left(S[n+1] - \hat{S}[n+1]\right)\right\} \\
 &= E\left\{S^2[n+1]\right\} - E\left\{S[n+1]\hat{S}[n+1]\right\} \\
 &= \frac{\sigma_A^2}{2} - \mathbf{c}_{XS}^T \mathbf{C}_X^{-1} E\{S[n+1]\mathbf{X}\} = \frac{\sigma_A^2}{2} - \mathbf{c}_{XS}^T \mathbf{C}_X^{-1} \mathbf{c}_{XS} \\
 &= \frac{\sigma_A^2}{2} - \frac{\sigma_A^2}{2} \begin{bmatrix} \cos(2\pi f_0) \\ \cos(4\pi f_0) \end{bmatrix}^T \begin{bmatrix} \frac{\sigma_A^2}{2} + \sigma_w^2 & \frac{\sigma_A^2}{2} \cos(2\pi f_0) \\ \frac{\sigma_A^2}{2} \cos(2\pi f_0) & \frac{\sigma_A^2}{2} + \sigma_w^2 \end{bmatrix}^{-1} \cdot \frac{\sigma_A^2}{2} \begin{bmatrix} \cos(2\pi f_0) \\ \cos(4\pi f_0) \end{bmatrix} \\
 &= \frac{\sigma_A^2}{2} \left(1 - \begin{bmatrix} a_1 & a_2 \end{bmatrix} \begin{bmatrix} \cos(2\pi f_0) \\ \cos(4\pi f_0) \end{bmatrix}\right) \\
 &= \frac{\sigma_A^2}{2} (1 - a_1 \cos(2\pi f_0) - a_2 \cos(4\pi f_0))
 \end{aligned}$$

where

$$a_1 = \gamma \cdot \frac{(1+\gamma)\cos(2\pi f_0) - \gamma\cos(4\pi f_0)\cos(2\pi f_0)}{(1+\gamma)^2 - \gamma^2 \cos^2(2\pi f_0)}$$

$$a_2 = -\gamma \cdot \frac{\gamma\cos^2(2\pi f_0) - (1+\gamma)\cos(4\pi f_0)}{(1+\gamma)^2 - \gamma^2 \cos^2(2\pi f_0)}$$

When $\gamma \rightarrow 0$ the random quantity to be estimated $S[n+1]$ is uncorrelated with the observed data $X[n]$ and $X[n-1]$. As a consequence, $a_1 \rightarrow 0$, $a_2 \rightarrow 0$ and so $\hat{S}[n+1] \rightarrow 0$, which is the a priori mean of $S[n+1]$.

When $\gamma \rightarrow \infty$, $X[n] = S[n]$, hence we go back to the case of 1-step pure prediction of a predictable process:

$$\begin{aligned}
 \lim_{\gamma \rightarrow \infty} a_1 &= \lim_{\gamma \rightarrow \infty} \left[\gamma \cdot \frac{(1+\gamma)\cos(2\pi f_0) - \gamma\cos(4\pi f_0)\cos(2\pi f_0)}{(1+\gamma)^2 - \gamma^2 \cos^2(2\pi f_0)} \right] = \frac{\cos(2\pi f_0) - \cos(4\pi f_0)\cos(2\pi f_0)}{1 - \cos^2(2\pi f_0)} \\
 &= 2\cos(2\pi f_0)
 \end{aligned}$$

$$\lim_{\gamma \rightarrow \infty} a_2 = \lim_{\gamma \rightarrow \infty} \left[-\gamma \cdot \frac{\gamma\cos^2(2\pi f_0) - (1+\gamma)\cos(4\pi f_0)}{(1+\gamma)^2 - \gamma^2 \cos^2(2\pi f_0)} \right] = -\frac{\cos^2(2\pi f_0) - \cos(4\pi f_0)}{1 - \cos^2(2\pi f_0)} = -1$$



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$$\lim_{\gamma \rightarrow \infty} \hat{S}[n+1] = 2 \cos(2\pi f_0) X[n] - X[n-1] = 2 \cos(2\pi f_0) S[n] - S[n-1]$$

$$\begin{aligned} \lim_{\gamma \rightarrow \infty} MSE\{\hat{S}[n+1]\} &= \lim_{\gamma \rightarrow \infty} \frac{\sigma_A^2}{2} (1 - h[1] \cos(2\pi f_0) - h[2] \cos(4\pi f_0)) = \\ &= \frac{\sigma_A^2}{2} (1 - 2 \cos(2\pi f_0) \cos(2\pi f_0) + \cos(4\pi f_0)) = 0 \end{aligned}$$

The LMMSE estimator of $S[n+1]$ and its MSE when $\sigma_A^2 = 1$, $\gamma = 2$, and $f_0 = 1/6$:

$$\begin{aligned} \hat{S}[n+1] &= 2 \cdot \frac{3 \cos(\pi/3) - 2 \cos(2\pi/3) \cos(\pi/3)}{9 - 4 \cos^2(\pi/3)} X[n] - 2 \cdot \frac{2 \cos^2(\pi/3) - 3 \cos(2\pi/3)}{9 - 4 \cos^2(\pi/3)} X[n-1] \\ &= 0.5 X[n] - 0.5 X[n-1] \end{aligned}$$

$$MSE\{\hat{S}[n+1]\} = \frac{1}{2} [1 - 0.5 \cos(\pi/3) + 0.5 \cos(2\pi/3)] = 0.25$$